PAPER ID-310851

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM V) THEORY EXAMINATION 2025-26
STRUCTURAL ANALYSIS

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

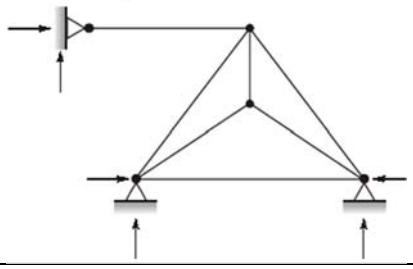
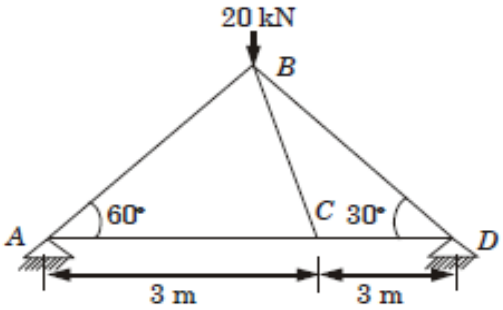
SECTION A

1. Attempt *all* questions in brief. **2 x 07 = 14**

Q no.	Question	CO	Level
a.	Define static indeterminacy of a beam.	1	K1
b.	What is a tension coefficient?	2	K1
c.	Write Castigliano's first theorem.	3	K1
d.	What is meant by absolute maximum bending moment?	4	K1
e.	Mention one application of influence line diagrams.	4	K2
f.	Name any two types of arches.	5	K1
g.	State Eddy's theorem.	5	K2

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 07**

a.	Explain the concept of static indeterminacy of structures and find the degree of static indeterminacy for the 2-D truss shown in the figure. 	1	K4
b.	Determine the forces in all members of the truss as shown in Fig. 	2	K4
c.	Determine the slope and deflection at the free end of a cantilever beam of span L subjected to a point load W at the free end using conjugate beam method. Take EI as constant.	3	K4
d.	Three-wheel loads 30 kN, 90 kN and 90 kN spaced 6 m apart from each other, with the 30 kN in the lead, pass over a simply supported beam of span 30 m. Determine the absolute maximum shear force and moment. Consider that loading can move in either direction.	4	K4
e.	For a three hinged arch discuss the procedure to draw influence lines for Normal thrust, Radial shear, Horizontal thrust & Bending moment.	5	K2

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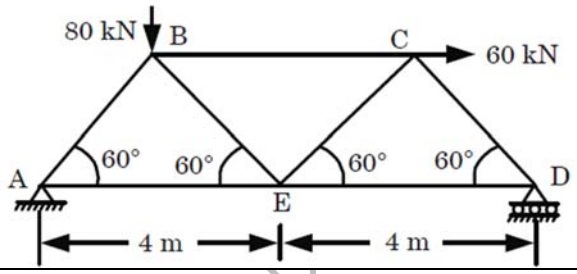
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SECTION C

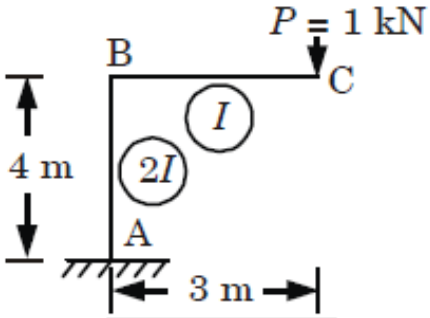
3. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Explain the load transfer mechanism in beams, trusses & frames.	1	K2
b.	A foot bridge is carried over a river of span 100 m. The supports are 4m and 12m higher than the lowest point of the cable. Determine the length of the cable. If the horizontal deck is loaded by UDL of 30 kN/m, find the tension in the cable.	1	K4

4. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Explain the procedure of method of section in truss analysis.	2	K2
b.	Analyze the truss shown in Fig. by the method of tension coefficient and determine the forces in all the members. 	2	K5

5. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Determine the vertical deflection at point C in the frame shown in Fig. Given E = 200 kN/mm2 and I = 30 × 106 mm4. 	3	K4
b.	Using the principle of strain energy, prove Maxwell's reciprocal theorem and clearly state the assumptions involved.	3	K4

6. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Draw the schematic ILD for reaction, SF and BM at a section when a unit load moves over a simply supported beam.	4	K3
b.	State the propositions for several concentrated loads moving over a simply supported beam. Explain and prove any one proposition.	4	K3

7. Attempt any *one* part of the following: **07 x 1 = 07**

a.	A three hinged parabolic arch of 60 m span and a rise of 10 m is subjected to a uniformly distributed load of 30 kN/m intensity over its left half portion and point load of 120 kN at right quarter span. Calculate	5	K4

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	the bending moment, normal thrust and radial shear at a section 10 m from the left support.		
b.	A three hinged circular arch of span 140 m and a rise of 35 m is subjected to a uniformly distributed load of 20 kN/m over the left half span. Determine the maximum bending moment in the arch. Also, calculate the normal thrust and radial shear at left quarter span.	5	K4

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